ASSIGNMENT-2

> [07/07/2026: RETREIVING THE DATA USING SQL SELECT STATEMENT]

1. Determine the Structure of Dept Table and its Contents.

```
Command Prompt - sqlplus /

SQL> create table dept(
  2  deptno int primary key,
  3  dname varchar2(14),
  4  location varchar2(13)
  5  );

Table created.

SQL> insert into dept values(10,'ACCOUNTING','NEW YORK');

1 row created.

SQL> insert into dept values(20,'RESEARCH','DALLAS');

1 row created.

SQL> insert into dept values(30,'SALES','CHICAGO');

1 row created.

SQL> insert into dept values(40,'OPERATIONS','BOSTON');

1 row created.

SQL> commit;

Commit complete.

SQL> desc dept;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DEPTNO                                    NOT NULL NUMBER(38)
 DNAME                                              VARCHAR2(14)
 LOCATION                                           VARCHAR2(13)

SQL> select * from dept;

    DEPTNO DNAME          LOCATION
---------- -------------- -------------
        10 ACCOUNTING     NEW YORK
        20 RESEARCH       DALLAS
        30 SALES          CHICAGO
        40 OPERATIONS     BOSTON
```

2. Determine the Structure of Emp Table and its Contents.

```
SQL> create table emp(
  2  empno int primary key,
  3  ename varchar2(10),
  4  job varchar2(10),
  5  salary int,
  6  deptno int
  7  );

Table created.

SQL> insert into emp values(7369,'SMITH','CLERK',800,20);

1 row created.

SQL> insert into emp values(7499,'ALLEN','SALESMAN',1600,30);

1 row created.

SQL> insert into emp values(7566,'ARUN','MANAGER',2975,20);

1 row created.

SQL> insert into emp values(7788,'KIRAN','ANALYST',3000,20);

1 row created.

SQL> insert into emp values(7839,',MEGHA','PRESIDENT',5000,10);

1 row created.

SQL> desc emp;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                              VARCHAR2(10)
 JOB                                                VARCHAR2(10)
 SALARY                                             NUMBER(38)
 DEPTNO                                             NUMBER(38)

SQL> select * from emp;

     EMPNO ENAME      JOB            SALARY     DEPTNO
---------- ---------- ---------- ---------- ----------
      7369 SMITH      CLERK             800         20
      7499 ALLEN      SALESMAN         1600         30
      7566 ARUN       MANAGER          2975         20
      7788 KIRAN      ANALYST          3000         20
      7839 ,MEGHA     PRESIDENT        5000         10
```

3. Display the Ename and Deptno from Emp table whose Empno is 7788.

```
SQL> select ename,deptno
  2  from emp
  3  where empno=7788;

ENAME          DEPTNO
---------- ----------
KIRAN              20

SQL>
```

ASSIGNMENT-3

>[08/07/2026:RESTRICTING AND SORTING DATA]

```
SQL> create table emp1
  2  (
  3  empno int,
  4  ename varchar(20),
  5  job varchar(20),
  6  mgr int,
  7  hiredate date,
  8  sal int,
  9  comm int,
 10  deptno int
 11  );

Table created.

SQL> insert into emp1 values(7369,'smitha','clerk',7902,'17-dec-80',800,null,20);

1 row created.

SQL> insert into emp1 values(7499,'allen','salesman',7698,'20-feb-81',1600,300,30);

1 row created.

SQL> insert into emp1 values(7566,'jones','manager',7839,'02-apr-81',2975,null,20);

1 row created.

SQL> insert into emp1 values(7698,'blake','manager',7839,'01-may-81',2850,null,30);

1 row created.

SQL> insert into emp1 values(7788,'scott','analyst',7566,'19-apr-87',3000,null,20);

1 row created.

SQL> insert into emp1 values(7839,'king','president',null,'17-nov-81',5000,null,10);

1 row created.

SQL> insert into emp1 values(7844,'tarun','salesman',7698,'08-sep-81',1500,0,30);

1 row created.

SQL> insert into emp1 values(7900,'james','clerk',7698,'03-dec-81',950,null,30);

1 row created.

SQL> insert into emp1 values(7934,'miller','clerk',7782,'23-jan-82',1300,null,10);

1 row created.
```

1. Display the Ename, Sal, Comm from Emp table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause.

```
SQL> select ename, sal, comm
  2  from emp1
  3  where comm is not null
  4  order by 2 desc, 3 desc;

ENAME                       SAL       COMM
-------------------- ---------- ----------
allen                      1600        300
tarun                      1500          0
```

2. The HR department needs a query to display all unique job codes from the EMP table.

```
SQL> select distinct job
  2  from emp1;

JOB
--------------------
clerk
salesman
manager
analyst
president
```

3.The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Alias.

```
SQL> select empno "emp #",
  2         ename "employee",
  3         job "job",
  4         hiredate "hire date"
  5  from emp1;

     emp # employee             job                  hire date
---------- -------------------- -------------------- ---------
      7369 smitha               clerk                17-DEC-80
      7499 allen                salesman             20-FEB-81
      7566 jones                manager              02-APR-81
      7698 blake                manager              01-MAY-81
      7788 scott                analyst              19-APR-87
      7839 king                 president            17-NOV-81
      7844 tarun                salesman             08-SEP-81
      7900 james                clerk                03-DEC-81
      7934 miller               clerk                23-JAN-82

9 rows selected.
```

4. The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the Job ID (separated by a comma and a space) and name the column Employee and Title by giving Column Alias.

```
SQL> select ename || ', ' || job "employee and title"
  2  from emp1;

employee and title
------------------------------------------
smitha, clerk
allen, salesman
jones, manager
blake, manager
scott, analyst
king, president
tarun, salesman
james, clerk
miller, clerk

9 rows selected.
```

5. Create a query to display all the data from EMP table. Separate each column output by a comma. Ename, Job, Hiredate, Mgr. Name the column title THE_OUTPUT.

```
SQL> select ename || ',' || job || ',' || hiredate || ',' || mgr "the_output"
  2  from emp1;

the_output
--------------------------------------------------------------------------------
smitha,clerk,17-DEC-80,7902
allen,salesman,20-FEB-81,7698
jones,manager,02-APR-81,7839
blake,manager,01-MAY-81,7839
scott,analyst,19-APR-87,7566
king,president,17-NOV-81,
tarun,salesman,08-SEP-81,7698
james,clerk,03-DEC-81,7698
miller,clerk,23-JAN-82,7782

9 rows selected.
```

6. Create a report to display the ename, job, and Hiredate for the employees whose name is SCOTT or TURNER. Order the query in ascending order by hiredate.

```
SQL> select ename, job, hiredate
  2  from emp1
  3  where ename in ('scott','turner')
  4  order by hiredate;

ENAME                JOB                  HIREDATE
-------------------- -------------------- ---------
turner               salesman             08-SEP-81
scott                analyst              19-APR-87
```

7. Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename.

```
SQL> select ename, deptno
  2  from emp1
  3  where deptno in (20,30)
  4  order by ename;

ENAME                    DEPTNO
-------------------- ----------
allen                        30
blake                        30
james                        30
jones                        20
scott                        20
smitha                       20
tarun                        30

7 rows selected.
```

8.Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary.

```
SQL> select ename "employee",
  2         sal "monthly salary"
  3  from emp1
  4  where sal between 2000 and 3000
  5  and deptno in (20,30);

employee             monthly salary
-------------------- --------------
jones                          2975
blake                          2850
scott                          3000
```

9. The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981.

```
SQL> select ename, hiredate
  2  from emp1
  3  where hiredate between '01-jan-1981' and '31-dec-1981';

ENAME                HIREDATE
-------------------- ---------
allen                20-FEB-81
jones                02-APR-81
blake                01-MAY-81
king                 17-NOV-81
tarun                08-SEP-81
james                03-DEC-81

6 rows selected.
```

10. Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.

```
SQL> select ename, sal
  2  from emp1
  3  where sal > &salary;
Enter value for salary: 2000
old   3: where sal > &salary
new   3: where sal > 2000

ENAME                       SAL
-------------------- ----------
jones                      2975
blake                      2850
scott                      3000
king                       5000
```

11. Create a report to display the last name and job title of all employees who do not have a manager.

```
SQL> select ename, job
  2  from emp1
  3  where mgr is null;

ENAME                JOB
-------------------- --------------------
king                 president
```

12. Create a query that prompts the user for Manager ID and generate EMPNO, ENAME, SAL, DEPTNO. The user should have the ability to sort the records on a selected Column.

```
SQL> select empno, ename, sal, deptno
  2  from emp1
  3  where mgr = &mgr_id
  4  order by &column_name;
Enter value for mgr_id: 7698
old   3: where mgr = &mgr_id
new   3: where mgr = 7698
Enter value for column_name: ename
old   4: order by &column_name
new   4: order by ename

     EMPNO ENAME                       SAL     DEPTNO
---------- -------------------- ---------- ----------
      7499 allen                      1600         30
      7900 james                       950         30
      7844 tarun                      1500         30
```

13. Create a query that prompts the user for a MGR and generates the empno, ename, salary, and department for that manager's employees. The HR department wants the ability to sort the report on a selected column.

```
SQL> select empno, ename, sal, deptno
  2  from emp1
  3  where mgr = &mgr
  4  order by &column_name;
Enter value for mgr: 7698
old   3: where mgr = &mgr
new   3: where mgr = 7698
Enter value for column_name: ename
old   4: order by &column_name
new   4: order by ename

     EMPNO ENAME                       SAL     DEPTNO
---------- -------------------- ---------- ----------
      7499 allen                      1600         30
      7900 james                       950         30
      7844 tarun                      1500         30
```

14.Display all employee last names in which the third letter of the name is A.

```
SQL> select ename
  2  from emp1
  3  where ename like '__a%';

ENAME
--------------------
blake
```

15. Display the last name of all employees who have both an A and an S in their ename.

```
SQL> select ename
  2  from emp1
  3  where ename like '%a%'
  4  and ename like '%s%';

ENAME
--------------------
smitha
james
```

16. Display the Ename, Job, Sal for all employees whose jobs are CLERK and whose salary is 800 or 950 or 1300.

```
SQL> select ename, job, sal
  2  from emp1
  3  where job = 'clerk'
  4  and sal in (800,950,1300);

ENAME                JOB                         SAL
-------------------- -------------------- ----------
smitha               clerk                       800
james                clerk                       950
miller               clerk                      1300
```

ASSIGNMENT-4

> [09/07/2026:USING SINGLE ROW FUNCTIONS TO CUSTOMIZE THE OUTPUT]

1.Write a query to display the current date. Label the column **Date**.

```
SQL> select sysdate "date"
  2  from dual;

date
---------
11-JUL-26
```

2.The HR department needs a report to display the employee number, last name, salary, and salary increased by **15.5%** (expressed as a whole number) for each employee. Label the column **New Salary**.

```
SQL> select empno,
  2         ename,
  3         sal,
  4         round(sal*1.155) "new salary"
  5  from emp1;

     EMPNO ENAME                       SAL new salary
---------- -------------------- ---------- ----------
      7369 smith                       800        924
      7499 allen                      1600       1848
      7566 jones                      2975       3436
      7698 blake                      2850       3292
      7788 scott                      3000       3465
      7839 king                       5000       5775
      7844 turner                     1500       1733
      7900 james                       950       1097
      7934 miller                     1300       1502

9 rows selected.
```

3. Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column **Increase**.

```
SQL> select empno,
  2         ename,
  3         sal,
  4         round(sal*1.155) "new salary",
  5         round(sal*1.155)-sal "increase"
  6  from emp1;

     EMPNO ENAME                       SAL new salary   increase
---------- -------------------- ---------- ---------- ----------
      7369 smith                       800        924        124
      7499 allen                      1600       1848        248
      7566 jones                      2975       3436        461
      7698 blake                      2850       3292        442
      7788 scott                      3000       3465        465
      7839 king                       5000       5775        775
      7844 turner                     1500       1733        233
      7900 james                       950       1097        147
      7934 miller                     1300       1502        202

9 rows selected.
```

4. Write a query that displays the **ename** (with the first letter uppercase and all other letters lowercase) and the length of the **ename** for all employees whose ename starts with the letters **J, A, or M**. Give each column an appropriate label. Sort the results by the employees' enames.

```
SQL> select initcap(ename) "employee",
  2         length(ename) "length"
  3  from emp1
  4  where ename like 'j%'
  5  or ename like 'a%'
  6  or ename like 'm%'
  7  order by ename;

employee                 length
-------------------- ----------
Allen                         5
James                         5
Jones                         5
Miller                        6
```

5. Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters **H** when prompted for a letter, then the output should show all employees whose last name starts with the letter **H**.

```
SQL> select ename
  2  from emp1
  3  where ename like '&letter%';
Enter value for letter: a
old   3: where ename like '&letter%'
new   3: where ename like 'a%'

ENAME
--------------------
allen
```

6. The HR department wants to find the length of employment for each employee. For each employee, display the **ename** and calculate the number of months between today and the date on which the employee was hired. Label the column **MONTHS_WORKED**. Order your results by the number of months employed. Round the number of months up to the closest whole number.

```
SQL> select ename,
  2         round(months_between(sysdate,hiredate)) "months_worked"
  3  from emp1
  4  order by round(months_between(sysdate,hiredate));

ENAME                months_worked
-------------------- -------------
scott                          471
miller                         534
james                          535
king                           536
turner                         538
blake                          542
jones                          543
allen                          545
smith                          547

9 rows selected.
```

7. Create a report that produces the following for each employee: **earns monthly but wants <3 times salary>**. Label the column **Dream Salaries**.

```
SQL> select ename||' earns '||sal||' monthly but wants '||sal*3 "dream salaries"
  2  from emp1;

dream salaries
--------------------------------------------------------------------------------
smith earns 800 monthly but wants 2400
allen earns 1600 monthly but wants 4800
jones earns 2975 monthly but wants 8925
blake earns 2850 monthly but wants 8550
scott earns 3000 monthly but wants 9000
king earns 5000 monthly but wants 15000
turner earns 1500 monthly but wants 4500
james earns 950 monthly but wants 2850
miller earns 1300 monthly but wants 3900

9 rows selected.
```

8. **.** Create a query to display the last name and salary for all employees. Format the salary to be **15 characters long, left-padded with the $ symbol**. Label the column **SALARY**.

```
SQL> select ename,
  2         lpad(sal,15,'$') "salary"
  3  from emp1;

ENAME
--------------------
salary
------------------------------------------------------------
smith
$$$$$$$$$$$$800

allen
$$$$$$$$$$$1600

jones
$$$$$$$$$$$2975

ENAME
--------------------
salary
------------------------------------------------------------
blake
$$$$$$$$$$$2850

scott
$$$$$$$$$$$3000

king
$$$$$$$$$$$5000

ENAME
--------------------
salary
------------------------------------------------------------
turner
$$$$$$$$$$$1500

james
$$$$$$$$$$$$950

miller
$$$$$$$$$$$1300

9 rows selected.
```

9. Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column **REVIEW**. Format the dates to appear in the format similar to **"Monday, the Thirty-First of July, 2000."**

```
SQL> select ename,
  2         hiredate,
  3         to_char(next_day(add_months(hiredate,6),'monday'),
  4         'fmday," the "ddsp" of "month, yyyy') "review"
  5  from emp1;

ENAME                HIREDATE
-------------------- ---------
review
--------------------------------------------------------------------------------
smith                17-DEC-80
monday, the twenty-two of june, 1981

allen                20-FEB-81
monday, the twenty-four of august, 1981

jones                02-APR-81
monday, the five of october, 1981


ENAME                HIREDATE
-------------------- ---------
review
--------------------------------------------------------------------------------
blake                01-MAY-81
monday, the two of november, 1981

scott                19-APR-87
monday, the twenty-six of october, 1987

king                 17-NOV-81
monday, the twenty-four of may, 1982


ENAME                HIREDATE
-------------------- ---------
review
--------------------------------------------------------------------------------
turner               08-SEP-81
monday, the fifteen of march, 1982

james                03-DEC-81
monday, the seven of june, 1982

miller               23-JAN-82
monday, the twenty-six of july, 1982
```

10. Display the last name, hire date, and day of the week on which the employee started. Label the column **DAY**. Order the results by the day of the week, starting with Monday.

```
SQL> select ename,
  2         hiredate,
  3         to_char(hiredate,'day') "day"
  4  from emp1
  5  order by to_char(hiredate,'d');

ENAME                HIREDATE  day
-------------------- --------- ------------------------------------
scott                19-APR-87 sunday
king                 17-NOV-81 tuesday
turner               08-SEP-81 tuesday
smith                17-DEC-80 wednesday
jones                02-APR-81 thursday
james                03-DEC-81 thursday
allen                20-FEB-81 friday
blake                01-MAY-81 friday
miller               23-JAN-82 saturday

9 rows selected.
```

11. Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show **"No Commission."** Label the column **COMM**.

```
SQL> select ename,
  2         nvl(to_char(comm),'No Commission') "comm"
  3  from emp1;

ENAME                comm
-------------------- ----------------------------------------
smith                No Commission
allen                300
jones                No Commission
blake                No Commission
scott                No Commission
king                 No Commission
turner               0
james                No Commission
miller               No Commission

9 rows selected.
```

12. **.** Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column **EMPLOYEES AND THEIR SALARIES**.

```
SQL> select rpad(substr(ename,1,8),8)||' '||
  2         rpad('*',sal/1000,'*') "employees and their salaries"
  3  from emp1
  4  order by sal desc;

employees and their salaries
--------------------------------------------------------------------------------
king     *****
scott    ***
jones    **
blake    **
allen    *
turner   *
miller   *
james
smith

9 rows selected.
```

13. Using the **DECODE** function, write a query that displays the grade of all employees based on the value of the column **JOB**, using the following data:

- PRESIDENT-A
- MANAGER-B
- ANALYST-C
- SALESMAN-C
- CLERK-D

```
SQL> select ename,
  2         job,
  3         decode(job,
  4         'president','A',
  5         'manager','B',
  6         'analyst','C',
  7         'salesman','C',
  8         'clerk','D') "grade"
  9  from emp1;

ENAME                JOB                  g
-------------------- -------------------- -
smith                clerk                D
allen                salesman             C
jones                manager              B
blake                manager              B
scott                analyst              C
king                 president            A
turner               salesman             C
james                clerk                D
miller               clerk                D

9 rows selected.
```

14. Rewrite the statement in the preceding exercise using the **CASE** syntax.

```
SQL> select ename,
  2         job,
  3         case
  4              when job='president' then 'A'
  5              when job='manager' then 'B'
  6              when job='analyst' then 'C'
  7              when job='salesman' then 'C'
  8              when job='clerk' then 'D'
  9         end "grade"
 10  from emp1;

ENAME                JOB                  g
-------------------- -------------------- -
smith                clerk                D
allen                salesman             C
jones                manager              B
blake                manager              B
scott                analyst              C
king                 president            A
turner               salesman             C
james                clerk                D
miller               clerk                D

9 rows selected.
```

ASSIGNMENT-5

10/07/2026: REPORTING AGGREGATED DATA USING THE GROUP FUNCTIONS

No.	Cat	Hands-on Assignment	Topics Covered	Status
1	M	Find the highest, lowest, sum, and average salary of all employees. Label the columns	Group Functions	☐
2	M	Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.	Group Functions	☐
3	M	Modify the above query to display the minimum, maximum, sum, and average salary for each job type.	Group by Clause	☐
4	M	Write a query to display the number of people with the same job	Group Functions	☐
5	M	Determine the number of managers without listing them. Label the column Number of Managers	Group by Clause	☐
6	M	Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.	Group Functions	☐
7	M	Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is $2000 or less. Sort the output in descending order of salary.	Where Clause, Group by Clause, Having Clause, Order by Clause	☐
8	M	Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.	Single Row and Group Functions	☐

```
SQL> select max(sal) as "maximum",
  2         min(sal) as "minimum",
  3         sum(sal) as "sum",
  4         round(avg(sal)) as "average"
  5  from emp1;

   maximum    minimum        sum    average
---------- ---------- ---------- ----------
      5000        800      19975       2219

SQL> select job,
  2         min(sal) as "minimum",
  3         max(sal) as "maximum",
  4         sum(sal) as "sum",
  5         round(avg(sal)) as "average"
  6  from emp1
  7  group by job;

JOB                     minimum    maximum        sum    average
-------------------- ---------- ---------- ---------- ----------
clerk                       800       1300       3050       1017
salesman                   1500       1600       3100       1550
manager                    2850       2975       5825       2913
analyst                    3000       3000       3000       3000
president                  5000       5000       5000       5000

SQL> select job,
  2         count(*) as "number of people"
  3  from emp1
  4  group by job;

JOB                  number of people
-------------------- ----------------
clerk                               3
salesman                            2
manager                             2
analyst                             1
president                           1

SQL> select count(distinct mgr) as "number of managers"
  2  from emp1;

number of managers
------------------
                 5
```

```
SQL> select max(sal)-min(sal) as "difference"
  2  from emp1;

difference
----------
      4200

SQL> select mgr,
  2         min(sal) as "lowest salary"
  3  from emp1
  4  where mgr is not null
  5  group by mgr
  6  having min(sal) > 2000
  7  order by min(sal) desc;

       MGR lowest salary
---------- -------------
      7566          3000
      7839          2850

SQL> select count(*) as "total employees",
  2         sum(case when extract(year from hiredate)=1980 then 1 else 0 end) as "1980",
  3         sum(case when extract(year from hiredate)=1981 then 1 else 0 end) as "1981",
  4         sum(case when extract(year from hiredate)=1982 then 1 else 0 end) as "1982"
  5  from emp1;

total employees       1980       1981       1982
--------------- ---------- ---------- ----------
              9          1          6          1

SQL> |
```

ASSIGNMENT-6

No.	Cat	Hands-on Assignment
1	M	Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.
2	M	Using set operator display the DEPTNO,SUM(SAL) for each dept, JOB,SUM(SAL) for each Job and Total Salary.
3	M	Using Set Operator display the JOB and Deptno in employees working in deptno 20,10,30 in that order.

```
SQL> select job,
  2         sum(case when deptno=10 then sal else 0 end) as dept10,
  3         sum(case when deptno=20 then sal else 0 end) as dept20,
  4         sum(case when deptno=30 then sal else 0 end) as dept30,
  5         sum(sal) as total
  6  from emp1
  7  group by job;

JOB                      DEPT10     DEPT20     DEPT30      TOTAL
-------------------- ---------- ---------- ---------- ----------
clerk                      1300        800        950       3050
salesman                      0          0       3100       3100
manager                       0       2975       2850       5825
analyst                       0       3000          0       3000
president                  5000          0          0       5000
```

```
SQL> select to_char(deptno) as category,
  2         sum(sal) as total_salary
  3  from emp1
  4  group by deptno
  5  union
  6  select job as category,
  7         sum(sal) as total_salary
  8  from emp1
  9  group by job
 10  union
 11  select 'total' as category,
 12         sum(sal) as total_salary
 13  from emp1;

CATEGORY                                 TOTAL_SALARY
---------------------------------------- ------------
20                                               6775
30                                               6900
10                                               6300
clerk                                            3050
salesman                                         3100
manager                                          5825
analyst                                          3000
president                                        5000
total                                           19975

9 rows selected.
```

```
SQL> select job,deptno
  2  from emp1
  3  where deptno=20
  4  union
  5  select job,deptno
  6  from emp1
  7  where deptno=10
  8  union
  9  select job,deptno
 10  from emp1
 11  where deptno=30;

JOB                     DEPTNO
-------------------- ----------
clerk                       20
manager                     20
analyst                     20
president                   10
clerk                       10
salesman                    30
manager                     30
clerk                       30

8 rows selected.
```

ASSIGNMENT-7

1	M	Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME,SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.	Natural Join
2	M	The HR department needs a report of all employees. Write a query to display the JOB,MGR,SAL,COMM,DNAME of employees whose JOB is SALESMAN.	Equi Join
3	M	The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.	Equi Join
4	M	Create a report to display employees' ename and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.	Self Join
5	M	Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.	Outer Join
6	M	The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.	Non-Equi Join
7	M	Display the ENAME,DNAME of all the employees. Also display those department name which do not have any employees working.	Outer Join
8	M	The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.	Self Join with Outer Join
9	M	Display the EMPNO,ENAME,DNAME,LOC of those employees who are working as CLERK. Use the USING clause.	USING Clause
10	M	Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000. Use the ON clause.	On Clause
11	M	Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees. Use LEFT OUTER JOIN.	LEFT OUTER JOIN
12	M	Display the ENAME,DNAME of employees. Use RIGHT OUTER JOIN.	RIGHT OUTER JOIN
13	M	Display the EMPNO,DNAME,LOC of employees. Use FULL OUTER JOIN.	FULL OUTER JOIN

```
SQL> select empno,ename,sal,dname,location
  2  from emp1
  3  natural join dept;

     EMPNO ENAME                       SAL DNAME     LOCATION
---------- -------------------- ---------- -------------- -------------
      7369 smith                       800 RESEARCH  DALLAS
      7499 allen                      1600 SALES     CHICAGO
      7566 jones                      2975 RESEARCH  DALLAS
      7698 blake                      2850 SALES     CHICAGO
      7788 scott                      3000 RESEARCH  DALLAS
      7839 king                       5000 ACCOUNTING        NEW YORK
      7844 turner                     1500 SALES     CHICAGO
      7900 james                       950 SALES     CHICAGO
      7934 miller                     1300 ACCOUNTING        NEW YORK

9 rows selected.

SQL> select job,mgr,sal,comm,dname
  2  from emp1 e,dept d
  3  where e.deptno=d.deptno
  4  and job='salesman';

JOB                         MGR        SAL       COMM DNAME
-------------------- ---------- ---------- ---------- --------------
salesman                   7698       1600        300 SALES
salesman                   7698       1500          0 SALES

SQL> select ename,job,e.deptno,dname
  2  from emp1 e,dept d
  3  where e.deptno=d.deptno
  4  and location='DALLAS';

ENAME                JOB                      DEPTNO DNAME
-------------------- -------------------- ---------- --------------
smith                clerk                        20 RESEARCH
jones                manager                      20 RESEARCH
scott                analyst                      20 RESEARCH

SQL> select e.ename as employee,
  2         e.empno as "emp#",
  3         m.ename as manager,
  4         m.empno as "mgr#"
  5  from emp1 e,emp1 m
  6  where e.mgr=m.empno;

EMPLOYEE                   emp# MANAGER              mgr#
-------------------- ---------- -------------------- ----------
scott                      7788 jones                7566
allen                      7499 blake                7698
turner                     7844 blake                7698
james                      7900 blake                7698
jones                      7566 king                 7839
blake                      7698 king                 7839

6 rows selected.
```

```
SQL> select e.ename as employee,
  2         e.empno as "emp#",
  3         m.ename as manager,
  4         m.empno as "mgr#"
  5  from emp1 e
  6  left join emp1 m
  7  on e.mgr=m.empno
  8  order by e.empno;

EMPLOYEE                   emp# MANAGER              mgr#
-------------------- ---------- -------------------- ----------
smith                      7369
allen                      7499 blake                7698
jones                      7566 king                 7839
blake                      7698 king                 7839
scott                      7788 jones                7566
king                       7839
turner                     7844 blake                7698
james                      7900 blake                7698
miller                     7934

9 rows selected.

SQL> select ename,
  2         job,
  3         dname,
  4         sal,
  5         grade
  6  from emp1 e,salgrade s,dept d
  7  where e.deptno=d.deptno
  8  and sal between losal and hisal;

ENAME                JOB                  DNAME   SAL       GRADE
-------------------- -------------------- -------------- ---------- ----------
miller               clerk                ACCOUNTING           1300          2
king                 president            ACCOUNTING           5000          5
smith                clerk                RESEARCH800          1
jones                manager              RESEARCH       2975          4
scott                analyst              RESEARCH       3000          4
james                clerk                SALES   950          1
allen                salesman             SALES          1600          3
turner               salesman             SALES          1500          3
blake                manager              SALES          2850          4

9 rows selected.

SQL> select ename,dname
  2  from dept d
  3  left join emp1 e
  4  on d.deptno=e.deptno;

ENAME                DNAME
-------------------- --------------
smith                RESEARCH
allen                SALES
jones                RESEARCH
```

```
blake                SALES
scott                RESEARCH
king                 ACCOUNTING
turner               SALES
james                SALES
miller               ACCOUNTING
                     OPERATIONS

10 rows selected.

SQL> select e.ename,
  2         e.hiredate,
  3         m.ename,
  4         m.hiredate
  5  from emp1 e
  6  left join emp1 m
  7  on e.mgr=m.empno
  8  where e.hiredate<m.hiredate;

ENAME                HIREDATE  ENAME                HIREDATE
-------------------- --------- -------------------- ---------
allen                20-FEB-81 blake                01-MAY-81
jones                02-APR-81 king                 17-NOV-81
blake                01-MAY-81 king                 17-NOV-81

SQL> select empno,ename,dname,location
  2  from emp1
  3  join dept
  4  using(deptno)
  5  where job='clerk';

     EMPNO ENAME                DNAME          LOCATION
---------- -------------------- -------------- -------------
      7934 miller               ACCOUNTING     NEW YORK
      7369 smith                RESEARCH       DALLAS
      7900 james                SALES          CHICAGO

SQL> select ename,sal,mgr,dname
  2  from emp1 e
  3  join dept d
  4  on e.deptno=d.deptno
  5  where sal>2000;

ENAME                       SAL        MGR DNAME
-------------------- ---------- ---------- --------------
king                       5000            ACCOUNTING
jones                      2975       7839 RESEARCH
scott                      3000       7566 RESEARCH
blake                      2850       7839 SALES
```

```
SQL> select empno,ename,job,e.deptno,dname,location
  2  from emp1 e
  3  left outer join dept d
  4  on e.deptno=d.deptno;

     EMPNO ENAME                JOB                      DEPTNO DNAME
---------- -------------------- -------------------- ---------- -------------
LOCATION
-------------
      7839 king                 president                    10 ACCOUNTING
NEW YORK

      7934 miller               clerk                        10 ACCOUNTING
NEW YORK

      7369 smith                clerk                        20 RESEARCH
DALLAS


     EMPNO ENAME                JOB                      DEPTNO DNAME
---------- -------------------- -------------------- ---------- -------------
LOCATION
-------------
      7566 jones                manager                      20 RESEARCH
DALLAS

      7788 scott                analyst                      20 RESEARCH
DALLAS

      7499 allen                salesman                     30 SALES
CHICAGO


     EMPNO ENAME                JOB                      DEPTNO DNAME
---------- -------------------- -------------------- ---------- -------------
LOCATION
-------------
      7698 blake                manager                      30 SALES
CHICAGO

      7844 turner               salesman                     30 SALES
CHICAGO

      7900 james                clerk                        30 SALES
CHICAGO


9 rows selected.

SQL> select ename,dname
  2  from emp1 e
  3  right outer join dept d
  4  on e.deptno=d.deptno;

ENAME                DNAME
-------------------- --------------
smith                RESEARCH
allen                SALES
jones                RESEARCH
blake                SALES
scott                RESEARCH
king                 ACCOUNTING
turner               SALES
james                SALES
miller               ACCOUNTING
                     OPERATIONS

10 rows selected.
```

```
SQL> select empno,dname,location
  2  from emp1 e
  3  full outer join dept d
  4  on e.deptno=d.deptno;

     EMPNO DNAME          LOCATION
---------- -------------- -------------
      7369 RESEARCH       DALLAS
      7499 SALES          CHICAGO
      7566 RESEARCH       DALLAS
      7698 SALES          CHICAGO
      7788 RESEARCH       DALLAS
      7839 ACCOUNTING     NEW YORK
      7844 SALES          CHICAGO
      7900 SALES          CHICAGO
      7934 ACCOUNTING     NEW YORK
           OPERATIONS     BOSTON

10 rows selected.
```